

# Gayathri Rajesh

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## EDUCATION

### University of California San Diego

Master of Science in Computer Science and Engineering – GPA 4.00/4.00

San Diego, CA

Sept. 2025 – Dec. 2026

### National Institute of Technology Tiruchirappalli

Bachelor of Technology in Electrical and Electronics Engineering – GPA 3.95/4.00

Tiruchirappalli, India

Dec. 2021 – Jul. 2025

## EXPERIENCE

### Cobot – SWE Intern - AI, Navigation and Controls

Jun. 2026 – Sep. 2026

- Designed an imitation learning-based camera-LiDAR sensor fusion pipeline for end-to-end local navigation, enabling map-free localization without relying on pre-built maps.
- Built an automated data collection, training, and validation pipeline in Isaac Sim, collecting 500 episodes of simulated navigation data to benchmark model performance across environments.

### UC San Diego – Graduate Student Researcher (Prof Henrik Christensen)

Oct. 2025 – Dec. 2025

- Developed a general-purpose 3D annotation pipeline for autonomous driving datasets using depth foundation models (**MapAnything**, **Depth Anything 3**) to generate dense scene reconstructions.
- Benchmarked depth model accuracy on the **nuScenes** dataset by reconstructing **3D point clouds** from predicted depth maps and quantifying geometric error against ground-truth point clouds via **Chamfer Distance**.

### University of Southern California – Robotics Research Intern (Prof Daniel Seita)

2024 – Mar. 2025

- Developed robotic fabric manipulation pipeline (folding, smoothing) using pre-trained foundation models, achieving a **50%** boost in folding performance & extending smoothing tasks to non-square fabrics in **SoftGym** simulations.
- Benchmarked **multimodal LLMs** (GPT, Gemini, Qwen-VL, etc.) over a set of 100 multiple-choice questions to test semantic understanding of fabrics achieving a maximum accuracy of **63%** with OpenAI o1.
- Implemented precise hand-eye calibration between cameras and bimanual UR5 arms using **ROS Noetic** and **MoveIt**, improving accuracy and reliability of robotic execution. [Project Report](#)

## PROJECTS

### Autonomous Scooter Hardware | [Report](#)

Apr. 2026 – Jun. 2026

- Designed the hardware base for an autonomous scooter platform integrating camera, **radar**, and steering subsystems.
- Interfaced a GoPro MAX2 camera with a RUBIK Pi and implemented real-time **object detection** using **YOLO V8**.
- Designed and deployed **ROS 2** nodes to synchronize camera, radar, and steering data streams, recording **ROS Bags** for downstream calibration and processing.

### Visual Inertial SLAM | [Report](#)

Mar. 2026

- Implemented a **visual-inertial SLAM** pipeline using an **Extended Kalman Filter** for nonlinear state estimation of robot trajectory from fused IMU and camera measurements.
- Performed IMU-based **localization** using the EKF predict step and visual landmark **mapping** via the EKF update step.
- Estimated the mean and covariance of the **motion model** and **observation model** to produce smooth trajectories.

### LiDAR-based SLAM | [Report](#)

Feb. 2026

- Estimated robot trajectory via scan matching between consecutive **LiDAR** point clouds using **Iterative Closest Point**.
- Built a **2D occupancy grid map** from LiDAR measurements, updating each cell's log-odds occupancy probability.
- Refined trajectory estimates using **Pose Graph Optimization** with **Levenberg-Marquardt** in **GTSAM** to correct accumulated drift.

### SLAM-based Roomba System | [Report](#)

Sep. 2025 – Dec. 2025

- Designed an autonomous Roomba system using **ROS 2 Jazzy** on a **RUBIK Pi**, achieving 90.1% area coverage.
- Implemented **EKF-based visual SLAM** for simultaneous **localization and mapping**, using AprilTag landmark detection for robust pose correction.
- Planned collision-free paths using the **A\* algorithm** and built a **Probabilistic Roadmap (PRM)** to identify safe trajectories.

### Unmanned Underwater Vehicle for Inspection and Repair | [Report](#)

Sep. 2023 – Feb. 2024

- Built a 5-DOF underwater robot with open-frame architecture, integrating cameras, IMU, and pressure sensors for real-time crack and corrosion inspection, with data relayed to on-ground stations.
- Applied transfer learning on VGG16 (**PyTorch**) for corrosion detection (**96.6% accuracy**) and developed computer vision models for crack detection, deploying them on a **Jetson Nano** for efficient onboard inference.

## TECHNICAL SKILLS

**Programming Languages:** Python | C/C++ | Matlab | Arduino C

**Libraries and Tools:** PyTorch | OpenCV | NumPy | Pandas | Matplotlib | Keras | ROS2 | SQL | Linux | Git

**Programmable Boards:** Arduino | Raspberry Pi | Jetson Nano | Rubik Pi